Física 3 – Semana 01: Termometria – Resolução

Prof. Vogt

1. C

Tomando por base o zero absoluto (0 K), vamos determinar seu valor correspondente nas demais escalas:

Celsius

$\theta\,(°C) = T(K) - 273 \Rightarrow \theta_C = 0 - 273$

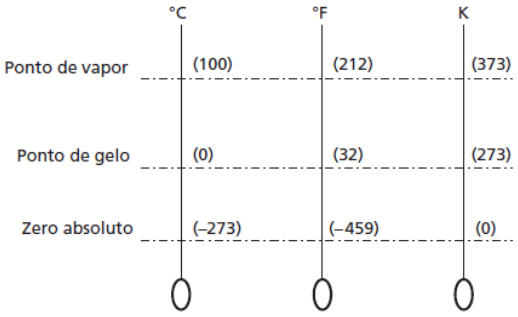
$\boxed{\theta_C = -273\ °C}$

Fahrenheit

$$\frac{\theta_F - 32}{9} = \frac{T - 273}{5} \Rightarrow \frac{\theta_F - 32}{9} = \frac{0 - 273}{5}$$

$\boxed{\theta_F \simeq -459\ °F}$

Observação: Para o aluno visualizar melhor, faça no quadro-de-giz o seguinte esquema:



2. B

$T = \theta_c + 273$

$T = 45 + 273$

$\boxed{T = 318\ K}$

$$\frac{\theta_c}{5} = \frac{\theta_F - 32}{9}$$

$$\frac{45}{5} = \frac{\theta_F - 32}{9}$$

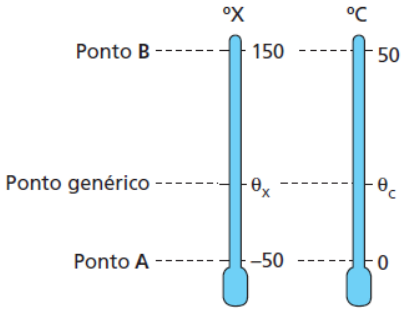
$$9 = \frac{\theta_F - 32}{9}$$

$\theta_F - 32 = 81$

$\boxed{\theta_F = 113°F}$

3.

a) Fazendo o esquema e relacionando as escalas **X** e Celsius, temos:



Do esquema, concluímos:

$$\frac{\theta_X - (-50)}{150 - (-50)} = \frac{\theta_C - 0}{50 - 0}$$

$$\frac{\theta_X + 50}{200} = \frac{\theta_C}{50} \Rightarrow \frac{\theta_X + 50}{4} = \theta_C$$

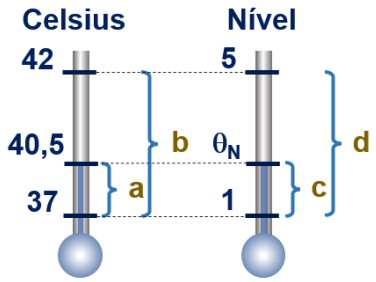
$$\theta_X + 50 = 4\theta_C \Rightarrow \boxed{\theta_X = 4\theta_C - 50}$$

b) Substituindo 80 °C na equação de conversão encontrada no item **a**, obtemos o θ_X correspondente:

$$\theta_X = 4(80) - 50 \Rightarrow \theta_X = 320 - 50$$

$$\boxed{\theta_X = 270\ °X}$$

4. E



$$\frac{a}{b} = \frac{c}{d}$$

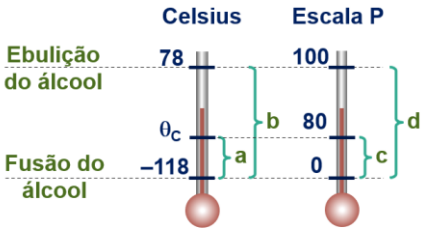
$$\frac{(40,5) - (37)}{(42) - (37)} = \frac{(\theta_N) - (1)}{(5) - (1)}$$

$$\frac{3,5}{5} = \frac{\theta_N - 1}{4}$$

$\theta_N - 1 = 2,8$

$\boxed{\theta_N = 3,8}$

5.



$$\frac{a}{b} = \frac{c}{d}$$

$$\frac{\theta_C - (-118)}{78 - (-118)} = \frac{80 - 0}{100 - 0}$$

$$\frac{\theta_C + 118}{196} = \frac{80}{100}$$

$10.\theta_C + 1180 = 1568$

$10.\theta_C = 388$

$\boxed{\theta_C = 38,8°C}$

6. B

$\Delta\theta = \theta - \theta_0$

$\Delta\theta = 80 - 25$

$\boxed{\Delta\theta = 55\ °C}$

$\Delta T = \Delta\theta_C$

$\boxed{\Delta T = 55\ K}$

$\Delta\theta_F = 1,8 \cdot \Delta\theta_C$

$\Delta\theta_F = 1,8\,(55)$

$\boxed{\Delta\theta_F = 99°F}$

7. C

Do enunciado temos que

$\theta_F = 2\theta_C + 2$

$$\frac{\theta_C}{5} = \frac{\theta_F - 32}{9}$$

$$\frac{\theta_C}{5} = \frac{(2\theta_C + 2) - 32}{9}$$

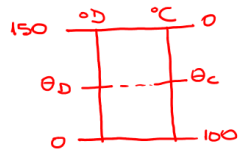
$$\frac{\theta_C}{5} = \frac{2\theta_C - 30}{9}$$

$$9\theta_C = 10\theta_C - 150$$

$$-10\theta_C + 9\theta_C = -150$$

$$\boxed{\theta_C = 150°C}$$

8. D



$$\frac{\theta_D - 0}{150 - 0} = \frac{\theta_C - 100}{0 - 100}$$

$$\frac{\theta_D}{\cancel{150}\ 3} = \frac{\theta_C - 100}{\cancel{-100}\ 2}$$

$$-2\theta_D = 3\theta_C - 300$$

$$-2\theta_D - 3\theta_C = -300 \quad \times(-1)$$

$$\boxed{2\theta_D + 3\theta_C = 300}$$